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Neural Differential Equations + Stochastic Calculus + Finance — Year 2 Roadmap

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🔵 Oct 12 – Oct 31 | Differential Equations Refresher & Study Setup

- **Objectives**
  • Refresh deterministic ODEs and numerical schemes.
  • Establish a structured note-taking system and learning framework.

- **Reading**
  • *MIT 18.03* Lectures 1–4 and exercises.[basically Imperial Year 1 Term 2 Calculus & Applications]
  • *Evans* intro sections (pre-Ch 1).
  • *Dive into Deep Learning* Ch. 4 (“Multilayer Perceptrons”), optionally skim Ch. 5 for implementation details.

- **Coding**
  • Implement Euler & RK4 methods for $\dot{x}=-x+t$.
  • Plot log–log error vs Δt and confirm expected orders (1 and 4).

- **Outreach / Logistics**
  • Create Notion or binder with sections (ODEs / Stochastic / Neural DE / Finance).
  • Draft 3-sentence introduction paragraph for faculty emails.
  • Save your convergence plot for potential use in outreach.

- **Milestones**
  ☑ ODE derivations complete ☑ Workspace ready ☑ Email draft prepared

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🔵 Nov 1 – Nov 15 | Brownian Motion & Stochastic Integration

- **Objectives**
  • Develop clear intuition for Brownian motion, martingales, and Ito integration.

- **Reading**
  • *Evans* Ch 1–2.
  • *Øksendal* §3–4.

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  • *Särkkä & Solin* §3.1–3.2.

- **Coding**
  • Simulate 10 Brownian paths (Δt = 0.01).
  • Numerically verify $E[W_t^2] \approx t$.

- **Outreach / Logistics**
  • Send Wave 1 faculty emails (6–9 recipients) with Brownian path figure + summary of Kidger (2022) survey on Neural DE families.

- **Milestones**
  ☑ Evans Ch 1–2 complete ☑ One-page summary of concepts ☑ Simulation verified ☑ First outreach sent

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🔵 Nov 16 – Nov 30 | Ito Calculus & Ito's Lemma

- **Objectives**
  • Understand and prove Ito's Lemma.
  • Apply it to geometric Brownian motion (GBM).

- **Reading**
  • *Evans* Ch 3.
  • *Shreve Vol II* §3.1–3.3.

- **Coding**
  • Implement GBM simulation (\mu = 0.05, \sigma = 0.2).
  • Overlay empirical $E[S_t]$ with analytical curve.

- **Outreach / Logistics**
  • Send Wave 2 emails (remaining faculty) with GBM plot and short progress note.

- **Milestones**
  ☑ Ito's Lemma proof complete ☑ GBM verified ☑ Outreach finished

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🔵 Dec 1 – Dec 15 | SDE Theory & Numerics

- **Objectives**
  • Study existence/uniqueness and numerical solution methods for SDEs.

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- **Reading**
  • *Evans* Ch 4–5.
  • *Øksendal* §5–6.
  • *Särkkä & Solin* Ch 4.

- **Coding**
  • Implement Euler–Maruyama and Milstein on Ornstein–Uhlenbeck process $dX=-\theta Xdt+\sigma dW_t$.
  • Plot RMSE vs step size.

- **Milestones**
  ☑ Convergence plot complete ☑ Error order confirmed

🔵 Dec 16 – Dec 31 | Financial SDEs & Black–Scholes

- **Objectives**
  • Link Ito calculus to financial modelling and derivative pricing.

- **Reading**
  • *Shreve Vol II* §4–5 (Black–Scholes PDE).
  • *Brigo et al.* (2024) *Stochastic Liquidity as Proxy …* intro + model equations.

- **Coding (GitHub)**
  • Monte Carlo pricing of a European call option; confirm within 1% of analytic price.

- **Milestones**
  ☑ Black–Scholes derivation finished ☑ Pricing code validated

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🔵 Jan 1 – Jan 15 | Neural ODE Mathematics

- **Objectives**
  • Grasp continuous-depth networks and adjoint sensitivity method.

- **Reading**
  • *Chen et al.* (2018) §§2–4.
  • *Kidger* (2022) survey §§2–3.

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- **Coding (GitHub)**
  • Run `torchdiffeq` spiral demo; annotate each solver step and plot trajectories.

- **Milestones**
  ☑ Adjoint derivation done ☑ Annotated notebook saved

🔵 Jan 16 – Jan 31 | Neural SDEs

- **Objectives**
  • Connect stochastic calculus foundations to Neural SDE architectures.

- **Reading**
  • *Li et al.* (2020) §§1–3.
  • *Särkkä & Solin* §5.1–5.3.

- **Coding (GitHub)**
  • Train a small Neural SDE (MLP width 16) on synthetic GBM paths for 10 epochs using `torchsde`.
  • Plot learned vs true drift/volatility.

- **Milestones**
  ☑ NSDE training run ☑ Parameters visualised

🔵 Feb 1 – Feb 15 | Controlled Differential Equations (Neural CDEs)

- **Objectives**
  • Understand pathwise integration and rough-path motivation.

- **Reading**
  • *Kidger et al.* (2020) §§2–3.
  • *Friz & Hairer* Ch 1 overview.

- **Coding (GitHub)**
  • Run `torchcde` demo; modify latent size (16→64) to observe path smoothness and performance impact.

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- **Milestones**
  ☑ CDE form understood ☑ Demo executed

🔵 Feb 16 – Feb 29 | Stochastic Control & HJB

- **Objectives**
  • Learn stochastic control theory and derive Hamilton–Jacobi–Bellman PDEs.

- **Reading**
  • *Øksendal* §10–11.
  • *Muhle-Karbe* (2019) intro & methodology.

- **Coding (GitHub)**
  • Simulate wealth process under constant vs feedback control policies; compare expected utilities.

- **Milestones**
  ☑ HJB derivation recorded ☑ Control simulation plotted

🔵 Mar 1 – Mar 15 | FBSDEs & Volatility Structures

- **Objectives**
  • Connect forward–backward SDEs to PDE formulations and Deep BSDE solvers.

- **Reading**
  • *Evans* final chapters.
  • *Han et al.* (2018) §2.
  • *Brigo et al.* (2024) selected equations.

- **Coding (GitHub)**
  • Numerically verify FBSDE–PDE correspondence on 1D example.

- **Milestones**
  ☑ FBSDE mapping derived ☑ Summary note written

🔵 Mar 16 – Mar 31 | Probability Reinforcement

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- **Objectives**
  • Reinforce measure-theoretic concepts and martingale theorems.

- **Reading**
  • *Durrett* Ch 1–4.
  • Re-derive Dominated Convergence & Optional Stopping in your notes.

- **Coding**
  • None (focus on proofs and derivations).

- **Milestones**
  ☑ Probability glossary completed ☑ Core convergence theorems summarised

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🔵 Apr 1 – Apr 15 | Unified Neural DE Framework

- **Objectives**
  • Integrate ODE, SDE, and CDE frameworks conceptually and numerically.

- **Reading**
  • *Oh et al.* (2025) review paper.
  • *Kidger* (2022) sections on solver training and numerical stability.

- **Coding (GitHub)**
  • Build comparative table (ODE vs SDE vs CDE forms, solvers, data types).
  • Train Neural ODE vs Neural SDE on synthetic dataset; compare trajectory variance.

- **Milestones**
  ☑ Comparison table done ☑ Dual solver notebook saved

🔵 Apr 16 – Apr 30 | Literature Synthesis & Faculty Mapping

- **Objectives**
  • Map Imperial faculty research directions and align with your interests.

- **Reading**
  • One paper each from Salvi, Gonon, Muhle-Karbe, and Zhang.

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- **Coding**
  • None (analytical work only).

- **Milestones**
  ☑ Paper summaries written ☑ One-page memo drafted

🔵 May 1 – May 15 | Review & Consolidation

- **Objectives**
  • Review key stochastic calculus and finance results.

- **Reading**
  • Consolidate highlights from Evans, Øksendal, and Särkkä & Solin.

- **Coding (GitHub)**
  • Simulate Ornstein–Uhlenbeck SDE and verify analytical mean reversion property.

- **Milestones**
  ☑ Four-page summary sheet created ☑ OU verification done

🔵 May 16 – May 31 | Expository Mini Report

- **Objectives**
  • Write a concise 6-page expository report and slide summarising the year's learning.

- **Reading**
  • Review Chen 2018, Li 2020, Kidger 2020 for citation accuracy.

- **Coding**
  • Ensure all notebooks run cleanly; regenerate figures for the report.

- **Milestones**
  ☑ Report finalised ☑ Slide completed

🔵 Jun 1 – Jun 30 | Reflection & Outreach Preparation

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- **Objectives**
  • Synthesize insights and prepare for UROP discussions.

- **Reading**
  • *Salvi et al.* (2024) Neural SPDEs.
  • *Han et al.* (2018) revisited for further study ideas.

- **Coding**
  • Optional: extend Neural SDE to 2D system as prototype for UROP.

- **Milestones**
  ☑ Reflection page completed ☑ Outreach paragraph written ☑ Advanced papers read

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📚 Resource Summary **(this doesn't include ML resources in iPad Books such as Herrberg's and Goodfellow's)**

Category	Core Texts	Usage Window
Stochastic Calculus	Evans (Ch 1–6), Øksendal (Ch 3–11), Särkkä & Solin (Ch 3–5), Shreve Vol II	Oct–Feb
Probability Theory, ML	Durrett (Ch 1–4), *Dive into Deep Learning* (Zhang et al)	Oct, Mar
Neural DEs	Chen 2018 (ODEs), Li 2020 (SDEs), Kidger 2020 (CDEs), Oh 2025 (Review), Kidger 2022 (Survey), *Friz & Hairer* (CDEs)	Jan–Apr
Finance / Control	Brigo 2024, Muhle-Karbe 2019, Han 2018	Feb–Jun
Videos	MIT 18.03, Shreve CMU lectures, Kidger talks	Oct–Feb
Code Repos	`torchdiffeq`, `torchsde`, `torchcde`	Monthly

🎓 Deliverables Timeline

Date	Deliverable

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Nov 30	Ito calculus notes + GBM code
Jan 31	Neural ODE & SDE summaries
Mar 15	FBSDE mapping sheet
Apr 30	Faculty research memo
May 31	6-page expository report
Jun 30	Reflection + outreach paragraph

✅ Expected Outcome by June 2026

- Comprehensive binder/Notion covering ODE → Neural CDE.
- Fluency in Ito calculus, stochastic control, and financial SDEs.
- Readiness for UROP or Oxford MCF discussions.
- Two–three clean Jupyter notebooks verifying key theorems numerically.

Interesting theoretical side quests (taken from Kidger's talk):

1. Formulation of CDE (why f(y(s))dX(s) instead of f(y(s), X(s))ds) and connections to rough path theory
2. Log-ODE method and capacity to handle long time series
3. The strong theoretical guarantee of NCDEs as universal approximators
4. Choice of interpolation scheme in making a discrete (and irregular) time series continuous (in the context of CDEs)